

Learning Note on Empirical Processes

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In this note, I write down a few key definitions of tools relevant for empirical process theory¹. An empirical process is a stochastic process based on a random sample, and can be thought of as how the data-based estimate deviates from a true quantity. For example, the quantity $\sqrt{n}(F_n(t) - F(t))$ which denotes the empirical CDF deviation from the true CDF at point t is an empirical process indexed by t . This can be alternately rewritten as $\{\sqrt{n}(P_n f - P f) : f \in \mathcal{F}\}$ where $\mathcal{F} = \{\mathbb{1}\{x \leq t\}, t \in \mathbb{R}\}$.

Many of these definitions are useful primitives to study the convergence rates of nonparametric estimators, my motivation.

1 Some Basic Tools for Empirical Processes

Here, I mainly focus on definitions related to bracketing number, covering number, and VC dimension. For each, I provide at least one example of its use.

The first group of definitions relate to bracketing numbers. Intuitively, it is the minimum number of brackets needed to cover a function class, depending on the bracket size, function space, and the metric.

1.1 Bracketing Number

Definition 1.1 (Bracketing Number). *Let P be a probability measure on $\mathcal{X}$, and denote the space of functions $L_r(P)$ as all functions g on $\mathcal{X}$ with $(\int_{\mathcal{X}} g^r dP)^{1/r} < +\infty$. Also let $l, u \in L_r(P)$ with $P(l \leq u) = 1$. $f \in \mathcal{F}$ lies in an ϵ -bracket (l, u) if*

¹ I draw my learnings from [Kosorok \(2008\)](#) and [van der Vaart and Wellner \(1996\)](#).

$P(l \leq f \leq u) = 1$ and $\|l - u\|_{r,P} \leq \epsilon$. The **bracketing number** $N_{[]}(\epsilon, \mathcal{F}, L_r(P))$ is the minimum number of ϵ -brackets such that each $f \in \mathcal{F}$ lies in at least one bracket.

Example 1 (Monotone functions). Let P be the Uniform distribution on $\mathcal{X} = [0,1]$, and take the L^1 norm. Suppose $\mathcal{F}_0$ is the space of continuous functions on $\mathcal{X}$. The bracketing number is $N_{[]}(\epsilon, \mathcal{F}_0, L_1(P)) = \infty$ because a function here can be arbitrarily large. Now suppose instead we restrict to $\mathcal{F}_1$ the space of monotone (but not necessarily continuous) functions bounded by M . Then the bracketing number is of the order²

$$\log N_{[]}(\epsilon, \mathcal{F}_1, L_1(P)) \asymp M/\epsilon.$$

Proof. First, show that one can construct such a set of brackets. Let there be $k = \lceil M/\epsilon \rceil$ evenly-spaced (except at last) intervals at the points $0, \epsilon, 2\epsilon, \dots, M$ denoted by c_j , and let $a_0, a_1, \dots, a_k$ be the sequence where $a_j = \inf\{x : f(x) \geq c_j\}$. Then let l and u be defined such that $l = c_{j-1}, u = c_j$ on the sets $\{x : c_{j-1} \leq f(x) < c_j\}$. (l, u) are clearly ϵ -brackets. By basic combinatorics (stars and bars), choosing k increasing points from $k+1$ possible grid points is given by $\binom{2k}{k} \leq 4^k = \exp(O(M/\epsilon))$.

Next, we need to show that this is also a lower bound. This is slightly more involved, and one way is by the Gilbert-Shannon-Varshamov (GSV) bound. ■

Example 2 (Empirical CDF). Let P be a probability measure on $\mathcal{X} = \mathbb{R}$, take the L^1 norm, and suppose $\mathcal{F} = \{1\{x \leq t\}, t \in \mathbb{R}\}$. By a similar argument as the previous example, we can show that $N_{[]}(\epsilon, \mathcal{F}, L_1(P)) \leq O(1/\epsilon)$.

Definition 1.2 (Entropy with Bracketing). The **entropy with bracketing** is simply defined as the log of the bracketing number, $\log N_{[]}(\epsilon, \mathcal{F}, L_r(P))$.

Example 3 (Finite bracketing integral). Define the **bracketing integral** as

$$J_{[]}(\delta, \mathcal{F}, L_r(P)) \equiv \int_0^\delta \sqrt{\log N_{[]}(\epsilon, \mathcal{F}, L_r(P))} d\epsilon.$$

One Donsker theorem is that if $\mathcal{F}$ has finite bracketing integral with $r = 2$, then $\mathcal{F}$ is P -Donsker, which loosely means that the empirical process converges weakly.

²One can get the same result on the L^r norm, though the proof is more involved.

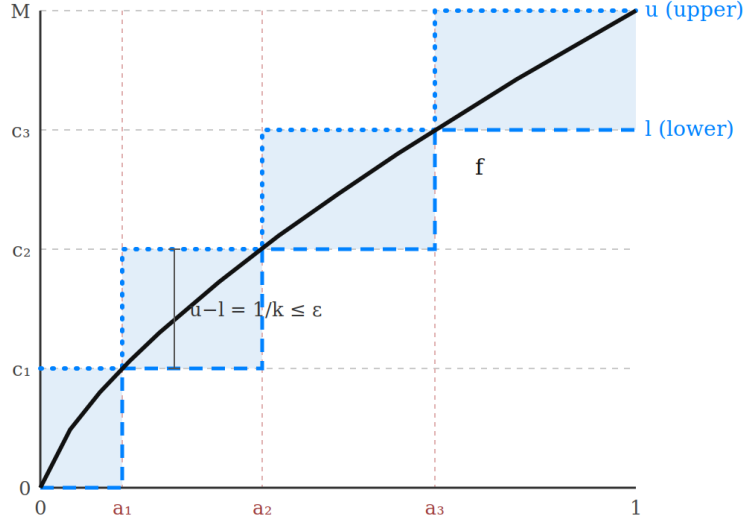


Figure 1: One possible bracketing construction for monotone functions.

1.2 Covering Number

The next group of definitions relate to covering number.

Definition 1.3 (Covering Number). *Fix a probability measure Q . The **covering number** $N(\epsilon, \mathcal{F}, L_r(Q))$ is the minimum number of ϵ -balls needed to cover $\mathcal{F}$, where an ϵ -ball around function $g \in L_r(Q)$ is the set $\{h \in L_r(Q) : \|h - g\|_{r,Q} \leq \epsilon\}$.*

The covering number is smaller than the bracketing number, i.e. $N(\epsilon, \mathcal{F}, L_r(P)) \leq N_{[]}(\epsilon, \mathcal{F}, L_r(P))$. This follows by using each optimal ϵ -bracket's lower function as the ϵ -cover ball center.

Example 4 (Monotone functions). It can be shown, perhaps unsurprisingly, that the covering number of monotone functions under the same setup as in Example 1 is again given by

$$\log N(\epsilon, \mathcal{F}_1, L_1(P)) \asymp (M/\epsilon).$$

Definition 1.4 (Uniform covering number and Uniform entropy integral). *Let $F : \mathcal{X} \rightarrow \mathbb{R}$ be an envelope of $\mathcal{F}$, i.e. it covers all $f \in \mathcal{F}$ over $\mathcal{X}$. The uniform covering number is given by*

$$\sup_{Q \text{ finitely discrete}} N(\epsilon \|F\|_{r,Q}, \mathcal{F}, L_r(Q)).$$

The uniform entropy integral is

$$J(\delta, \mathcal{F}, L_r) = \int_0^\delta \sqrt{\log \sup_Q N(\epsilon \|F\|_{r,Q}, \mathcal{F}, L_r(Q))} d\epsilon.$$

Example 5. A well known theorem (e.g. Thm 19.13, [van der Vaart \(1998\)](#)), loosely, is that if the uniform covering number is finite for all $\epsilon > 0$ and the outer measure of F is also finite, then $\mathcal{F}$ is P -Glivenko-Cantelli, which loosely is a strong form of convergence of the probability measure. Another well known theorem (e.g. Thm 19.14, [van der Vaart \(1998\)](#)) is that if the the uniform entropy integral is finite, and the outer measure of F^2 is also finite, then $\mathcal{F}$ is P -Donsker.

1.3 VC Dimension

The final group of definitions relate to the Vapnik-Chervonenkis (VC) dimension.

We begin by listing a few terminologies. Consider an arbitrary collection $\{x_1, \dots, x_n\}$ in a set $\mathcal{X}$ and a collection $\mathcal{C}$ of subsets of $\mathcal{X}$.

- $\mathcal{C}$ **picks out** a subset A of $\{x_1, \dots, x_n\}$ if $A = \{x_1, \dots, x_n\} \cap C$ for some $C \in \mathcal{C}$.
(intuitively: cut out exactly some intersected subset)
- $\mathcal{C}$ **shatters** $\{x_1, \dots, x_n\}$ if all 2^n subsets are picked out by $\mathcal{C}$.
- The **VC-index** of $\mathcal{C}$, $V(\mathcal{C})$, is the smallest n for which no set of size n $\{x_1, \dots, x_n\} \subset \mathcal{X}$ is shattered by $\mathcal{C}$. Furthermore, if $\mathcal{C}$ shatters all sets $\{x_1, \dots, x_n\}$ for $n \geq 1$, then its VC-index is ∞ .
- The **VC-dimension** is equal to VC-index - 1³. It is the largest n for which some set of size n is shattered.
- $\mathcal{C}$ is a **VC-class** if $V(\mathcal{C}) < \infty$.
- $\Delta_n(\mathcal{C}, x_1, \dots, x_n)$ is the number of subsets of $\{x_1, \dots, x_n\}$ that can be picked out with $\mathcal{C}$.

³It seems like the CS literature likes to use VC-dimension while the statistics literature more frequently uses VC-index.

Example 6 (VC dimension of points in the real line). Consider a set of points $\{x_1, \dots, x_n\}$ (WLOG, order them in ascending order) and let $\mathcal{C} = \{(-\infty, c) : c \in \mathbb{R}\}$. It is obvious that sets such as $\{x_2\}$, $\{x_2, x_3\}$ etc. cannot be picked out by $\mathcal{C}$ since it must also cover elements to the left of them. Hence, the VC-dimension is 1 and the VC-index is 2.

Example 7 (Infinite VC dimension). Consider a set of points on the real line $\{2^{-k} : k = 0, 1, 2, \dots\}$ and the collection $\mathcal{C} = \{x : \sin(\theta x) > 0\}$ over all $\theta > 0$. It can be shown that the VC index is $V(\mathcal{C}) = \infty$.

The following lemma states that for a VC-class, the number of subsets that can be picked out can increase in n in at most polynomial order.

Lemma 1 (Sauer's Lemma). *For a VC-class of sets $\mathcal{C}$,*

$$\max_{x_1, \dots, x_n \in \mathcal{X}} \Delta_n(\mathcal{C}, x_1, \dots, x_n) \leq \sum_{j=0}^{V(\mathcal{C})-1} \binom{n}{j}.$$

Note that for constant j , $\binom{n}{j} = O(n^j)$, thus the RHS is $O(n^{V(\mathcal{C})-1})$.

We can relate the covering number to the VC index as follows. Let

Definition 1.5 (VC-subgraph class). *A collection $\mathcal{F}$ of measurable functions is a VC-subgraph class of functions if the collection of all subgraph $\{(x, t) : f(x) < t\}$ forms a VC-class of sets.*

Intuitively, a VC-subgraph class is the functional analogue of VC-class, and hence is often simply called VC class. It intuitively means that the collection is not too rich in the subgraph space.

The following theorem further links the covering number of VC-classes of functions with the VC-index, and show that the covering number grows at polynomial rate just like for sets.

Theorem 1. *There exists a universal constant $K < \infty$ such that for any VC-subgraph class of measurable functions $\mathcal{F}$ with integrable envelope F , any $r \geq 1$, any probability measure Q with $\|F\|_{Q,r} > 0$, and any $0 < \epsilon < 1$,*

$$N(\epsilon \|F\|_{Q,r}, \mathcal{F}, L_r(Q)) \leq KV(\mathcal{F})(4e)^{V(\mathcal{F})} \left(\frac{2}{\epsilon}\right)^{r(V(\mathcal{F})-1)}.$$

Therefore, as discussed in example 5, VC subgraph classes of functions satisfy uniform entropy requirements of Glivenko-Cantelli and Donsker theorems.

Example 8 (Constant functions). Consider $\mathcal{X} = \mathbb{R}$ and the collection of constant functions $\mathcal{F} = \{f_c = c : c \in [0,1]\}$. The largest shatterable set is size 1, so the VC-index is $V(\mathcal{F}) = 2$ (the reasoning is just like Example 6, except that we now consider the half planes). Furthermore, by Theorem 1, the bound on the covering number is $N(\epsilon, \mathcal{F}, L_r(Q)) \leq 32Ke^2 \left(\frac{2}{\epsilon}\right)^r \asymp \epsilon^{-r}$.

References

- KOSOROK, M. R. (2008): *Introduction to Empirical Processes and Semiparametric Inference*, Springer Series in Statistics, New York: Springer.
- VAN DER VAART, A. W. (1998): *Asymptotic Statistics*, Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge: Cambridge University Press.
- VAN DER VAART, A. W. AND J. A. WELLNER (1996): *Weak Convergence and Empirical Processes: With Applications to Statistics*, Springer Series in Statistics, New York: Springer.